SPECTRAL BEHAVIOUR OF 2-PHENYLBENZOPYRYLIUM SALTS IN SOLUTIONS AS THE RESULT OF NUCLEOPHILIÑ SOLVATION

Alexander D. Roshal

Institute of Chemistry at Kharkov National University
4 Svobody sqr., Kharkov, 61077, Ukraine
Email: Alexandre.D.Rochal@univer.kharkov.ua

Dennis Yu. Minayev

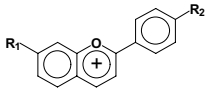
Physical Chemistry Laboratory South Parks Road
Oxford OX1 3QZ, Great Britain

Alexander I. Novikov

Department of Chemistry, Kharkov Military University,
6 Svobody sqr. 61022, Ukraine

INTRODUCTION

2-phenylbenzopyrylium salts (PBP) are widely used in different kinds of their practically important applications – from laser dyes to food industry. Various derivatives of PBP-salts – anthocyanins and anthocyanidins are known as natural plant dyes. Positions of long-wavelength absorption bands of newly synthesized phenylbenzopyrylium ions are used for the prognostic purposes and for prediction of their chemical hardness. Some PBP salts belong to the class of photochromic materials.



Spectral properties of PBP salts are very diverse. They change dramatically at environment properties changing.

The existing suppositions about changing of PBP salts spectral properties

HOWEVER

Solvatochromic and solvatofluorochromic effects depend on environment polarity

No correlation was found between absorption or emission band positions and solvent polarity functions

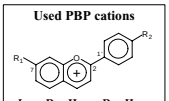
Twisted interfragmental charge transfer in the excited state

Interfragmental bond orders in several PBP cations increase in the excited state. In such case, twisted structure formation seems less favorable

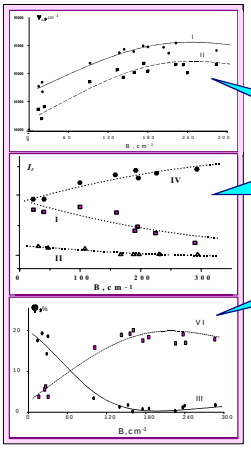
Influence of solvent nucleophilicity on spectral properties of PBP salts

Our investigations of several PBP perchlorates in 16 solvents allow us to suggest, that spectral parameters of these salts depend on the solvent nucleophilicity (B), empirical value characterizing the electron-donating ability of the solvent molecules.

Used PBP cations



I -	R_1 - H	R_2 - H
II -	R_1 - H	R_2 - OH
III -	R_1 - H	R_2 - OCH_3
IV -	R_1 - H	R_2 - Cl
V -	R_1 - H	R_2 - NO_2
VI -	R_1 - OH	R_2 - H
VII -	R_1 - OH	R_2 - OH



Correlation between positions of long-wavelength absorption bands and the solvent nucleophilicity

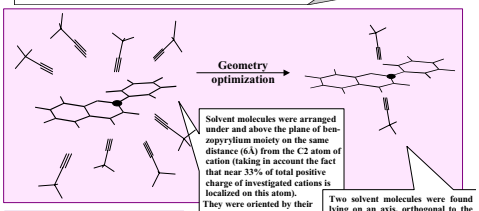
Correlation between positions of emission bands and the solvent nucleophilicity

Dependency of fluorescence quantum yield on the solvent nucleophilicity

All the presented plots have similar "saturation" regions, which corresponds to the most electron density redistribution on PBP cations under solvent influence

Simulation of nucleophilic solvation of PBP cations

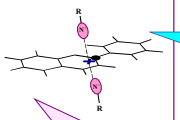
The "supermolecule" method was used for simulation of nucleophylic solvation. Optimisation of nucleophilic solvate complex geometry made done by AM1 method



Solvent molecules were arranged under and above the plane of benzopyrylium moiety on the same distance (6Å) from the C2 atom of cation (taking in account the fact that near 33% of total positive charge of investigated cations is localized on this atom). They were oriented by their nucleophilic centers towards the center of positive charge of cation.

Two solvent molecules were found lying on an axis, orthogonal to the plane of benzopyrylium cation on the distance of 3.0-3.5Å from the latter. Positions of other solvent molecules were left unchanged, some of the molecules even were moved off the cation on the distance up to 7-10Å.

A simple model of nucleophilic solvate complex



The model does not take into account influence of all solvent molecules in the solvate

It does not account macroscopic polar properties of solvent

It does not account H-bonding with solvent molecules

This simple model allow to evaluate the excitation re-solvation effect in PBP

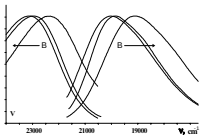
It allows to make calculations of electronic spectra of solvated PBP cations with the account of reasonable quantity of singly excited configurations

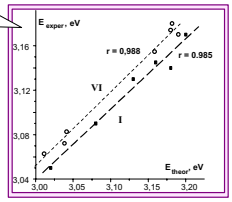
It allows to evaluate the influence of solvent molecules on the charge transfer in PBP cations at their electronic excitation

It allows also to estimate the spin-orbit coupling in the solvated cations.

Nucleophilic solvation and solvatochromic ability of PBP salts

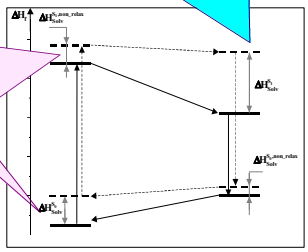
The model of nucleophilic solvate complex correctly describes the influence of solvent onto the spectral properties of PBP salts





Calculations of enthalpies of formation of solvated and non-solvated PBP cations demonstrate that the nucleophilic solvation leads to more significant stabilization of them in the ground state, in comparison with the Franc-Condon excited state. This phenomenon results in the bathochromic shifts of long-wavelength band in PBP salts absorption spectra in nucleophilic solvents.

The relaxation processes of solvated PBP-cations in the excited state leads to more significant decrease of enthalpie of formation of S_1-state, than that in the case of non-solvated cations. Consequently, the nucleophilic solvation results in bathochromic shifts of emission bands in fluorescence spectra of PBP cations.



Energy levels for non-solvated cations

Energy levels for solvated cations

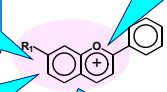
Spectral diversity of PBP-cations

Quantum chemical interpretation of the nature of electronic transitions in the absorption spectra of phenylbenzopyrylium salts and quantitative description of the excited state interfragmental charge transfer in them show that most of these salts could be classified onto two different groups, differs by the nature of their long-wavelength absorption and emission bands.

The first group consists of compounds, which have electron-donor substituents only in the benzopyrylium moiety.

The δ-order of interfragmental bond in the excited state exceeds that in the ground one

The low-energy electronic transition of these substances is localized mainly on benzopyrylium moiety



In the excited state PBP cations of the first group are more rigid in comparison with their ground state

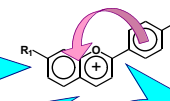
The excited state charge transfer between benzopyrylium and phenyl fragments in PBP cations is low.

Solvent nucleophilicity increase leads to the rise of their fluorescence quantum yield

The second group consists of cations, which have electron donor substituents in lateral phenyl cycle (independently from the nature and quantity of substituents in benzopyrylium moiety)

In the excited state PBP cations of the second group are less rigid in comparison with their ground state

The low-energy electronic transitions are characterized by the significant charge transfer from phenyl to benzopyrylium moiety



Solvent nucleophilicity increase leads to the rise of their fluorescence quantum yield

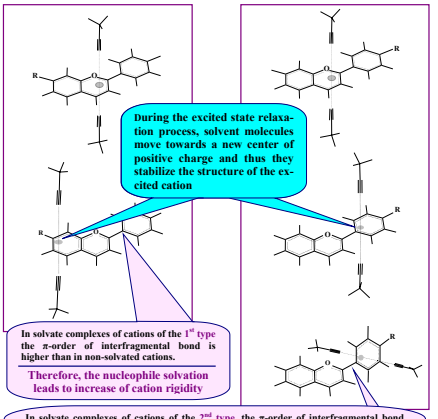
fragmental bond decreases at the cation excitation.

Structure of PBP solvate complexes in the excited state

Geometry of solvate complexes in the excited state depends on the type of PBP-cation

Cations of the first group

Cations of the second group



During the excited state relaxation process, solvent molecules move towards a new center of positive charge and thus they stabilize the structure of the excited cation

In solvate complexes of cations of the 1st type the π-order of interfragmental bond is higher than in non-solvated cations.

Therefore, the nucleophile solvation leads to increase of cation rigidity

In solvate complexes of cations of the 2nd type, the π-order of interfragmental bond decreases down to zero. Free rotation of the lateral phenyl ring around the interfragmental bond becomes possible.

Therefore, the nucleophile solvation leads to decrease of the rigidity of the cation of the 2nd type and to the twisted structure (TICT) formation.

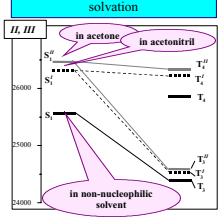
Influence of nucleophilic solvation on fluorescence quantum yields of PBP

We consider that the main channel of non-radiate deactivation of S_1-state of PBP cations and, consequently, the main cause of small values of fluorescence quantum yields is the effective intersystem crossing owing the increased spin-orbit interaction.

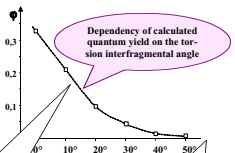
The most important reasons to spin-orbit coupling changing at nucleophilic solvation

Changing of energetic gap between S_1 and T_i levels at nucleophilic solvation

Changing of spin-orbit interaction efficiency due to decrease or increase of molecular rigidity



Decrease of molecular rigidity, rotation of fragments around a bond connecting them leads to formation of mixed states (ðð*, nð*, etc.) of different multiplicity, spin-orbit interaction between which is more significant than between ordinary ðð* singlet and triplet states.



Dependency of calculated quantum yield on the torsion interfragmental angle

Formation of nucleofilic complexes of PBP (*II*, *III*) in acetonitrile (B = 160 cm^{-1}) and in acetone (B = 224 cm^{-1}) can result in appearance a new channel of intersystem crossing S_1 ~ T_i, which leads to fluorescence quenching.

Nucleophilic solvation increases the rigidity of the cations of the 1st type **On going from non-nucleophilic to nucleophilic solvents the quantum yield of PBP cations increases**

Nucleophilic solvation decreases the rigidity of the cations of the 2nd type **On going from non-nucleophilic to nucleophilic solvents the quantum yield of these PBP cations decreases**

CONCLUSIONS

- Conception of the influence of nucleophilic solvation of phenylbenzopyrylium salts is proposed for clarification of their spectral properties.
- A simple model of nucleophilic solvate complex is proposed.
- The nucleophilic solvation stabilizes S_0-state and relaxed S_1-state of PBP cations. In the result, the increase of environment nucleophilicity leads to hypsochromic shift of long-wavelength absorption bands and bathochromic shift of emission band in PBP-salts electronic spectra.
- The nucleophilic solvation leads to the increase in the rigidity of PBP-cations, the electronic excitation of which is localized predominantly on the benzopyrylium fragment. In nucleophilic solvents the fluorescence quantum yield of these cations increases.
- The rigidity of PBP-cation, which S_1-state is characterized by significant interfragmental charge transfer, decreases under the influence of nucleophilic solvation. The quantum yield of corresponding PBP salts in nucleophilic environment decreases, too.